1st SIT EXAMINATION QUESTION PAPER: **Year Long 2022/23**

Module Code:	**CC6001NI**
Module Title:	**Advanced Database System Development**
Module Leader:	**Mr. Lekhnath Katuwal (Islington College)**

Date:	
Day / Evening:	**Day**
Start Time:	
Duration:	**2** Hours

Exam Type:	**Unseen, Closed**
Materials supplied:	**None**
Materials permitted:	Writing equipment only
Warning:	Candidates are warned that possession of unauthorised materials in an examination is a serious assessment offence.

Instructions to candidates:	**Please Note:** Inclusive of this cover page, this examination paper consists of **5** pages**.** The student must complete four (**4**) out of five (**5**) questions. This examination accounts for **40%** of your total module grades. You are to return this examination paper, BEFORE you leave the examination room. DO NOT TURN PAGE OVER UNTIL INSTRUCTED

Attempt Any Four Questions.

Question 1

1. a) Write Short Note on following topics

[2x5 = 10 Marks]

i) Cardinality

ii) Checkpoint

iii) Natural Join

iv) Group By Clause

1. b) Write the difference between Domain Integrity, Entity Integrity, and Referential Integrity.

[6 Marks]

1. c) The following relational schema specifies a part of a relational database for an employee information system:

Employee (<u>empid</u>, name, contact, salary, supervisor_id, department_id*)*
Department (<u>department_id</u>, name, location)

Write both SQL and Relational Algebra for the following

i) Find the name and salary of employees working on 'Finance' Department and employee name start with 'N'.

[2+2 = 4 Makrs]

ii) Find the name of employees with the name of their supervisor.

[2.5 + 2.5 = 5 Marks]

Question 2

2. a) In the course of transaction processing, the system takes checkpoints at specified intervals. Assume that sometime after a particular checkpoint was taken a system failure occurred. Consider the following transactions in the scenario:

> T1 started before the checkpoint and completed before the system failed.
> T2 started after the checkpoint and completed before the system failed.
> T3 started after the checkpoint but did not complete when the system failed.
> T4 started and completed before the checkpoint was taken.
> T5 started before the checkpoint but did not complete when the system failed.

(i) Draw a diagram to illustrate the above set of transactions.

[5 Marks]

(ii) For each of the above transactions, indicate which must be redone, which must be undone and which is unaffected by the failure.

[5 Marks]

2. b) Define Database Recovery. What are the techniques for Recovery? Explain each technique.

[2 Marks + 3 Marks + 4 Marks]

2. c) Explain the three level ANSI SPARC Database Architecture with diagram.

[6 Marks]

Question 3

3. a) Explain about Shared Lock and Exclusive Lock. Complete the following transaction with suitable application of Locks

[6 Marks + 4 Marks]

Time	T_A	T_B	Balance
			500
t1	begin		500
t2	read (balance)	begin	500
t3	balance=balance+200	read (balance)	500
t4	write (balance)	balance=balance-300	700
t5	commit	write (balance)	200
t6		commit	200

3. b) The following list represents the sequence of events in an interleaved execution of a set of transactions T1, T2, ... T24 in a concurrency system based on locking, where A, B, ...H are data items. Assume that FETCH A acquires an S lock on A, UPDATE A promotes that lock to an X lock, and all locks are held to the next synch point.

Time	***transaction***	***operation***
t0		
t1	T1	FETCH A
t2	T4	FETCH D
t3	T5	FETCH A
t4	T2	FETCH E
t5	T2	UPDATE E
t6	T3	FETCH F
t7	T2	FETCH F
t8	T1	COMMIT
t9	T6	FETCH A
t10	T5	ROLLBACK
t11	T6	FETCH C
t12	T6	UPDATE C
t13	T7	FETCH G
t14	T8	FETCH H
t15	T9	FETCH G
t16	T9	UPDATE G
t17	T8	FETCH E
t18	T7	COMMIT
t19	T9	FETCH H
t20	T3	FETCH G
t21	T10	FETCH E
t22	T9	UPDATE H
t23	T2	UPDATE F
t24		

(i) Provide a Wait-for-Graph to illustrate whether there is any deadlock at time *t24.* You are advised to provide full and appropriate workings.

[10 Marks]

(ii) Discuss how the system could recover if it were deadlocked at time *t24*, and justify your choice of victim transaction(s).

[5 Marks]

Question 4

4.a) There are several options available for distributing data in a distributed database system. Describe two options for data fragmentation (horizontal fragmentation and vertical fragmentation). For each option, give an example to illustrate your answer.

[9 Marks]

4.b) Define Normalization and Why Normalization is necessary?

[4 Marks]

4.c) Normalize the given table up to 3NF indicating all the dependency and Keys (primary and foreign) clearly.

Note: State any assumption that you have made during normalization.

[2 Marks + 3 Marks + 4 Marks + 3 Marks]

Student_ID	Name	Address	Age	Zip Code	City	Dept_ID	Department Name
2	Faiz	house 18 Defence Club	21	0	Karachi	4	Botany
2	Faiz	Street 92 house 2	21	48758	Lahore	4	Botany
3	Nouman	Street 19 House 20	19	9887	Faislabad	5	English
3	Nouman	Officer's Residence New York	19	6556	New York	5	English
4	Jerry	Street 18 House 29	22	266	Dubai	6	Physical Education
4	Jerry	Greens building 3	22	5555	Abu Dhabi	6	Physical Education

Question 5

5.a) An educational institution has several departments. There are various database objects in the system. You currently hold DBA privilege to the database.

i) Create two users as ‘DataQL’ and ‘DataDML’

[4 Marks]

ii) Grant all necessary privileges for a user ‘DataQL’ to only SELECT the MARKS table.

[3 Marks]

iii) Grant all necessary privileges for user ‘DataDML’ to ISNERT on STUDENT table.

[3 Marks]

iv) Grant all necessary privileges for user ‘DataDML’ to UPDATE, DELETE and SELECT on MODULE table.

[4 Makrs]

v) Write the command to take the privilege back from all users.

[4 marks]

5.b) Create an Access Matrix in tabular form from the above scenario before taking back all the privilege from the users.

[6 Marks]